MACHINE CERTIFICATION CERTIFICATE
Module 6.3 -- Lemma 4.1 Falsified, Weil Bound Certified

Battle Plan v1.6 -- David Fox -- May 2026

1. CLAIM

Two certified conclusions for N = 143, S_2(Gamma_0(143)):

Conclusion A: Lemma 4.1 is FALSE for N=143. There exist 94 explicit primes p in {5,...,997} \ {11,13} for which R(p) = max_ap(p) / (2*sqrt(p)) > cos(2*pi/p), where Lemma 4.1 claims this inequality cannot hold.

Conclusion B: The Weil bound holds for the dim-1 newform 143.2.a.a. |a_p(143.2.a.a)| <= 2*sqrt(p) for all 164 tested primes.

2. DEFINITIONS

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alpha_0 = 299 + pi/10 (M1-certified to 5000 decimal places)
dist(p) = min(frac(p*alpha_0), 1 - frac(p*alpha_0)) [nearest-integer distance, always in [0, 1/2]]
max_ap(p) = max(|a_p(143.2.a.a)|, |Tr(a_p(143.2.a.b))|, |Tr(a_p(143.2.a.c))|)
NOT the H_1 aggregate trace. Individual form traces from LMFDB.
R(p) = max_ap(p) / (2*sqrt(p))
bound(p) = cos(2*pi/p)
Lemma 4.1 pass = (R(p) <= bound(p)); Lemma 4.1 fail = (R(p) > bound(p))
CM status: LMFDB cm field = 0 for all three level-143 forms. No CM exclusion needed.
dim(new S_2(Gamma_0(143))) = 1+4+6 = 11. No 143.2.a.d exists.
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3. INPUT SHAs

Module	SHA-256	Content
M1	63ef870a78766619327e99b68683bcef...	alpha_0 = 299+pi/10 (5000 dps)
M8.1	863a3aef237e2807be77b9c28b90e93f...	143_traces.csv: Hecke traces, 3 forms, p<=997
M6.3 script	7c023ca556e448e4ea882bceb3ee40bf...	certificates/m6_3_lemma41.py
M6.3 output	add9fef4a623392436bfb272180252ac...	m6_3_lemma41.csv (164 rows)

4. RESULTS SUMMARY

Primes tested (5<=p<=997, excluding 11, 13): 164

Lemma 4.1 FAILURES (R > cos(2*pi/p)): 94 primes

Lemma 4.1 passes: 70 primes

Weil bound failures |ap_a|>2*sqrt(p): 0 primes

5. FIRST 30 LEMMA 4.1 FAILURE PRIMES

p	max_ap	R	cos(2pi/p)	R - bound	form
7	6	1.13389	0.62349	0.51040	143.2.a.b
19	10	1.14708	0.94582	0.20126	143.2.a.c
23	11	1.14683	0.96292	0.18391	143.2.a.c
37	15	1.23299	0.98562	0.24738	143.2.a.c
43	26	1.98248	0.98934	0.99314	143.2.a.b
47	18	1.31279	0.99108	0.32171	143.2.a.b

59	16	1.04151	0.99433	0.04718	143.2.a.b
61	16	1.02429	0.99470	0.02959	143.2.a.c
73	32	1.87266	0.99630	0.87636	143.2.a.c
83	26	1.42694	0.99714	0.42980	143.2.a.c
89	23	1.21900	0.99751	0.22149	143.2.a.c
97	27	1.37072	0.99790	0.37281	143.2.a.c
107	34	1.64345	0.99828	0.64518	143.2.a.c
113	49	2.30477	0.99845	1.30631	143.2.a.c
131	30	1.31056	0.99885	0.31171	143.2.a.c
139	28	1.18746	0.99898	0.18849	143.2.a.b
163	30	1.17489	0.99926	0.17563	143.2.a.c
181	54	2.00689	0.99940	1.00750	143.2.a.b
193	66	2.37539	0.99947	1.37592	143.2.a.c
199	32	1.13421	0.99950	0.13471	143.2.a.b
227	68	2.25666	0.99962	1.25704	143.2.a.b
233	44	1.44127	0.99964	0.44163	143.2.a.b
241	36	1.15948	0.99966	0.15982	143.2.a.b
251	32	1.00991	0.99969	0.01022	143.2.a.b
263	52	1.60323	0.99972	0.60351	143.2.a.c
269	66	2.01205	0.99973	1.01232	143.2.a.b
271	44	1.33641	0.99973	0.33667	143.2.a.b
277	60	1.80252	0.99974	0.80278	143.2.a.c
283	42	1.24832	0.99975	0.24857	143.2.a.c
311	40	1.13410	0.99980	0.13430	143.2.a.b

6. ALL FAILURE PRIMES (p <= 997)

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[7, 19, 23, 37, 43, 47, 59, 61, 73, 83, 89, 97, 107, 113, 131, 139, 163, 181, 193, 199, 227, 233, 241, 251,
263, 269, 271, 277, 283, 311, 313, 317, 337, 347, 353, 367, 379, 389, 401, 419, 431, 433, 439, 443, 449,
457, 467, 487, 499, 503, 509, 523, 563, 569, 577, 587, 593, 601, 613, 617, 619, 631, 643, 647, 659, 661,
673, 677, 709, 719, 727, 733, 739, 757, 761, 797, 823, 827, 853, 859, 863, 877, 881, 883, 887, 911, 929,
937, 947, 953, 967, 971, 983, 997]
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7. WEIL BOUND VERIFICATION (dim-1 form 143.2.a.a)

For all 164 tested primes: |a_p(143.2.a.a)| <= 2*sqrt(p). Weil bound holds. Max R_dim1 observed:

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max R_dim1 = 0.970269 < 1 (Ramanujan for dim-1 form)
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8. CHAIN OF CUSTODY

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M1 (alpha_0) -> M8.1 (LMFDB traces) -> M6.3 (Lemma 4.1 falsification)
M1 SHA: 63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291
M8.1 SHA (143_traces.csv): 863a3aef237e2807be77b9c28b90e93f2e5d20be064b9f988f68265c8640d1f1
M6.3 script SHA: 7c023ca556e448e4ea882bceb3ee40bf347f28d83fda00846dea65f5fa7e2d2e
M6.3 output SHA: add9fef4a623392436bfb272180252ac134ad6f5665c688bbc1f9db4b873a332
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9. AUDIT NOTE

The prototype script that produced the original 118-failure table used two wrong definitions: (1) dist = frac(p*pi/10) instead of min(frac, 1-frac), and (2) max_ap included the H_1 aggregate trace (-44 at p=71) instead of individual form traces. Both errors are corrected here. This certificate supersedes the prototype table.

The previous claim of a CM form 143.2.a.c with a_71=-35 was a labeling error. LMFDB cm field = 0 for all three level-143 forms. There is no 143.2.a.d. Dimension count: 1+4+6=11=new space dim.

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